Package 'CTTB'

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Type Package

Title Estimate Aggregated and Conditional Trimming Bounds Using Covariates

Version 0.1.0

Description Tools to estimate covariate-tightened trimming bounds for treatment effects with endogenous outcome missingness or sample selection. The methods extend Lee's trimming bounds by using generalized random forests to incorporate high-dimensional covariates, estimate conditional bounds, and aggregate them into tighter identified sets.

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Encoding UTF-8

Depends R (>= 3.5.0)

Imports ggplot2, grf, rlang

NeedsCompilation no

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`CTTB-package` *CTTB: Covariate-Tightened Trimming Bounds*

Description

Tools for estimating trimming bounds using covariates.

Details

The package provides functions for estimating classic Lee bounds, covariate-tightened trimming bounds, and conditional bounds under endogenous outcome missingness or sample selection.

`blattman_replic` *Replication Data Included with CTTB*

Description

Example dataset included in the package.

Usage

```
data(blattman_replic)
```

Format

An object included with the package.

Source

Included in the package source distribution.

`CTTB` *Covariate-tightened trimming bounds*

Description

Estimate aggregated and conditional trimming bounds using covariates.

Usage

```
CTTB(data, seed = NULL, Y, D, S, X = NULL, W = NULL, Pscore = NULL,
     splitting = FALSE, cv_fold = 5, trim_l = 0, trim_u = 1,
     aggBounds = TRUE, IM_cv = TRUE, alpha = 0.05, cBounds = FALSE,
     X_moderator = NULL, LeeBounds = FALSE, cond.mono = TRUE)
```

Arguments

`data`	A data frame.
`seed`	Optional random seed used for sample splitting and forest fitting.
`Y`	Name of the outcome variable.
`D`	Name of the binary treatment variable.
`S`	Name of the binary response or selection indicator.
`X`	Character vector of covariate names.
`W`	Optional name of a sampling-weight variable. If omitted, equal weights are used.
`Pscore`	Optional name of a treatment-probability variable. If omitted, propensity scores are estimated from the data.
`splitting`	Logical; whether to use regression splitting in quantile forests when applicable.
`cv_fold`	Number of folds used for cross-fitting.
`trim_l`	Lower trimming threshold applied to estimated probabilities.
`trim_u`	Upper trimming threshold applied to estimated probabilities.
`aggBounds`	Logical; whether to estimate aggregated covariate-tightened trimming bounds.
`IM_cv`	Logical; whether to report Imbens-Manski confidence intervals for the identified set.
`alpha`	Significance level used for confidence intervals.
`cBounds`	Logical; whether to estimate conditional bounds.
`X_moderator`	A data frame or matrix of moderator values at which conditional bounds are evaluated.
`LeeBounds`	Logical; whether to estimate classic trimming bounds.
`cond.mono`	Logical; whether to allow conditional monotonicity in the Lee-bounds step.

Details

The package implements covariate-tightened trimming bounds proposed by Samii, Wang, and Zhou (2025). The method extends Lee's trimming bounds by using random-forest estimates of nuisance quantities and cross-fitting to construct conditional and aggregated identified sets.

Value

A list of class `"CTTB"`. Depending on the chosen options, the list can contain `TB`, `CTTB`, `cB`, `parameters`, and `call`.

References

Athey, S., Tibshirani, J., and Wager, S. (2019). Generalized random forests. *Annals of Statistics*, 47(2), 1148–1178.

Lee, D. S. (2009). Training, wages, and sample selection: Estimating sharp bounds on treatment effects. *Review of Economic Studies*, 76(3), 1071–1102.

Samii, C., Wang, Y., and Zhou, J. A. (2025). Generalizing trimming bounds for endogenously missing outcome data using random forests. *Political Analysis*.

Examples

```
## Not run:
data(simData)
res <- CTTB(
  data = simData,
  Y = "Y", D = "D", S = "S",
  X = setdiff(names(simData), c("Y", "D", "S", "Ps")),
  Pscore = "Ps",
  aggBounds = TRUE,
  LeeBounds = TRUE
)
print(res)

## End(Not run)
```

`plot.CTTB` *Plot CTTB results*

Usage

```
## S3 method for class 'CTTB'
plot(x, y = NULL, type = "aggregated", moderator_plot = 1, ...)
```

Arguments

`x`	An object of class "CTTB".
`y`	Unused.
`type`	Either "aggregated" or "conditional".
`moderator_plot`	Index of the moderator to plot when type = "conditional".
`...`	Additional arguments, currently unused.

Value

A ggplot2 object.

Examples

```
## Not run:
data(simData)
res <- CTTB(simData, Y = "Y", D = "D", S = "S",
            X = setdiff(names(simData), c("Y", "D", "S", "Ps")),
            Pscore = "Ps", aggBounds = TRUE, LeeBounds = TRUE)
plot(res)

## End(Not run)
```

print.CTTB — *Print a CTTB object*

Usage

```
## S3 method for class 'CTTB'
print(x, ...)
```

Arguments

x	An object of class "CTTB".
...	Additional arguments, currently unused.

Examples

```
## Not run:
data(simData)
res <- CTTB(simData, Y = "Y", D = "D", S = "S",
            X = setdiff(names(simData), c("Y", "D", "S", "Ps")),
            Pscore = "Ps", aggBounds = TRUE, LeeBounds = TRUE)
print(res)

## End(Not run)
```

santoro_replic — *Replication Data Included with CTTB*

Description

Example dataset included in the package.

Usage

```
data(santoro_replic)
```

Format

An object included with the package.

Source

Included in the package source distribution.

`simData` *Simulated data objects bundled with CTTB*

Description

Simulation objects used in package examples and development. `simData` is the main example dataset. The package also includes `dat`, `dat_full`, and `true_parameters`.

Usage

```
data(simData)
```

Format

Objects included in the package source distribution.

Source

Included with the package.

`summary.CTTB` *Summarize a CTTB object*

Usage

```
## S3 method for class 'CTTB'
summary(object, ...)
```

Arguments

`object`	An object of class `"CTTB"`.
`...`	Additional arguments, currently unused.

Examples

```
## Not run:
data(simData)
res <- CTTB(simData, Y = "Y", D = "D", S = "S",
            X = setdiff(names(simData), c("Y", "D", "S", "Ps")),
            Pscore = "Ps", aggBounds = TRUE, LeeBounds = TRUE)
summary(res)

## End(Not run)
```

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